Marwan Amr

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Summary

Senior Data Scientist with production experience building multi-agent orchestration systems, time-series forecasting pipelines, and semantic search infrastructure. End-to-end project owner, from stakeholder problem scoping through database design, model development, performance optimization, and deployment, with proven business impact ($600K+ revenue uplift). Skilled in LLM-powered agentic workflows, MCP tool integration, full-stack performance tuning (PostgreSQL, Redis, React), and scalable MLOps.

Education

Zewail City, University of Science and Technology Giza, Egypt
Bachelor of Science in Data Science and Artificial Intelligence *Oct. 2022 – Jun. 2026*

- **Honors:** Awarded the Silver Medal at the graduation ceremony for winning 2nd place in Egypt's largest AI competition (Beltone AI Hackathon)
- **Relevant Coursework:** Machine Learning, Deep Learning, Computer Vision, NLP, Statistical Inference, Mathematical Optimization, Data Mining, Explainable AI, Algorithms, Databases

Awards

2nd Place – Beltone AI Hackathon | *Egypt's Largest AI Competition* 2025

- Won $4,000 for developing an AI-driven supply chain optimization solution achieving minimum-cost delivery with shortest time and 100% order fulfillment across complex logistics networks, competing against 3,500+ contributors, 1,000+ teams, and 34+ universities across Egypt

Experience

Senior Data Scientist Jun. 2026 – Present
Datalentech *Cairo, Egypt*
AI/ML Engineer *Jun. 2024 – Jun. 2026*

Multi-Agent AI Hiring Platform | *Python, FastAPI, PostgreSQL, Redis, Vertex AI, LangChain, pgvector, MCP*

- Owned end-to-end architecture of a multi-agent AI hiring platform, from stakeholder requirements through production deployment, orchestrating 5+ specialized agents for CV screening, skill analysis, candidate matching, interview generation, and career development
- Designed a multi-agent orchestration layer coordinating autonomous agents: a **Skills Recommendation Agent** that analyzes job descriptions to generate targeted skill sets per role, a **Question Generation Agent** that produces tailored interview questions given job requirements and candidate skills, and a **Career Development Agent** that leverages MCP server tool-calling to search real-time course databases and recommend personalized upskilling paths per candidate
- Built a **Pre-Interview Intelligence Agent** that aggregates multi-phase candidate scores (CV parsing, assessments, technical screens), synthesizes insights, and generates strategic talking points and validation questions for recruiters ahead of technical interviews
- Implemented semantic job-candidate matching using Vertex AI embeddings (3072-dim text-embedding-004) with pgvector cosine similarity search, achieving 85% candidate-job matching accuracy
- Designed Lambda/Beta ratio algorithm validating skill authenticity by comparing theoretical knowledge vs. practical experience signals extracted by LLM agents
- Built production-grade MLOps infrastructure with API rate limiting, lazy credential initialization, retry strategies, and quota management, improving candidate search throughput by 5×
- Designed and built the PostgreSQL database from scratch, including normalized schema design, table relationships, connection pooling, and migration workflows, then created pgvector-based IVFFlat indexing for high-dimensional embedding search across candidates and job listings
- Led a **multi-stage performance optimization initiative**: implemented database-level indexing and concurrency tuning, introduced Redis caching to eliminate redundant fetches, replaced sequential queries with batched parallel execution, optimized API endpoints with cursor-based pagination, restructured backend orchestration to minimize round-trips, and applied frontend caching with React.memo and memoization hooks to reduce unnecessary re-renders

Poultry Price Forecasting & Harvest Optimization | *Python, LightGBM, Optuna, SHAP, LIME*

- Owned the full project lifecycle, from problem scoping with business stakeholders through data pipeline design, model development, explainability analysis, and deployment, delivering an end-to-end forecasting system for 23+ SKUs across multiple product groups
- Trained LightGBM forecasting models with Optuna hyperparameter optimization on highly volatile sales data, engineering temporal, categorical, and cross-product features to improve predictive robustness
- Applied target normalization and signal stabilization transformations to handle extreme price volatility across market segments
- Built model explainability layer using SHAP, LIME, and custom feature degradation frameworks, enabling stakeholders to understand and trust forecast drivers
- Designed a harvest optimization plan driven by price forecasts that increased company profitability by 10%, equivalent to approximately $600,000 in additional revenue

Projects

Retrieval-Augmented Generation System | *Python, LangChain, DeepSeek*
- Built a RAG pipeline with custom retrievers for multi-source document ingestion (PDFs, codebases, web data)
- Implemented context-aware chunking and re-ranking to improve answer relevance and reduce hallucination

Multi-Person Tracking System | *Python, YOLOv12, OSNet, DeepSort*
- Implemented real-time multi-person tracking with re-identification embeddings for persistent identity across frames
- Improved tracking robustness under occlusion and re-entry conditions using appearance-based matching

Face Recognition Mobile App | *Swift, TensorFlow, ArcFace*
- Developed on-device face recognition iOS app using ArcFace embeddings, achieving 92% accuracy with cosine similarity verification

Explainable Credit Scoring System | *Python, XGBoost, LightGBM, SHAP, LIME*
- Built interpretable credit scoring models with SHAP, LIME, and counterfactual explanations for financial risk compliance

Technical Skills

Languages: Python, C/C++, C#, JavaScript, SQL, Swift, TypeScript
AI/ML: LightGBM, XGBoost, Time-Series Forecasting, Feature Engineering, Optuna, SHAP, LIME, Explainable AI
Agentic AI & LLMs: LangChain, MCP Servers, Tool-Calling Pipelines, Multi-Agent Orchestration, Google Gemini API
Frameworks: PyTorch, TensorFlow, Keras, Hugging Face, Scikit-learn, NLTK, OpenCV
Embeddings & Vector DB: Vertex AI Embeddings, pgvector, FAISS, ChromaDB, Cosine Similarity
Computer Vision: YOLO, DeepSort, OSNet, ArcFace
Backend & Databases: FastAPI, Flask, Node.js, PostgreSQL, Redis, Supabase, Connection Pooling, Cursor-Based Pagination
Frontend: React, React.memo, useMemo/useCallback, Client-Side Caching
DevOps & Cloud: Git, Docker, GCP, Vertex AI Platform, MLOps Pipelines